

Draft RSE Award

Categories, evaluation criteria and indicators

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Abstract

test abstract

Keywords: keyword1; keyword2; keyword3; keyword4

Abbreviations

RS	Research Software
DACH	Germany, Austria, Switzerland
GI	Gesellschaft für Informatik
BTW	Datenbanksysteme für Business, Technologie und Web
deRSE	Annual German Conference for Research Software Engineering

1 RSE Award categories

Category	Sub-Category	Format	Eligibility
(1) Scientific Excellence	(1.1) Humanities and Social Sciences	Open call	1. RS must be clearly associated with scientific peer-reviewed journal publications (substantial part of research) 2. Researchers must be affiliated with a German or DACH university/research institution 3. Only for (1.4) sub-category: RS must have an artefact badge from a German or DACH conference
	(1.2) Life Sciences		
	(1.3) Natural Sciences and Engineering		
	(1.4) Computer Science and Contributions with Artefacts		
(2) Newcomer	-	Open call	1. PhD (and master) students from a German or DACH university/research institution 2. Submission of motivation letter

Table 1: Award categories, format and eligibility criteria. RS = Research software

2 Process and timeline of RSE Award

2.1 Scientific Excellence category

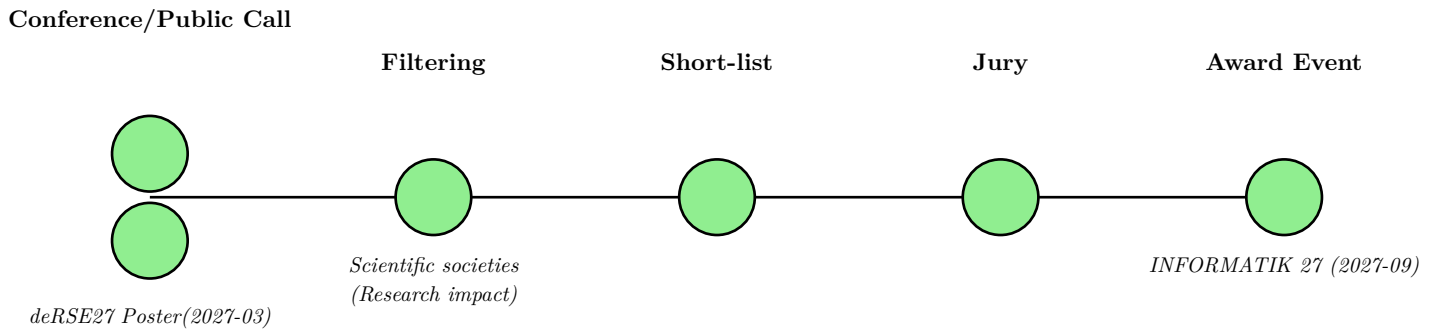


Figure 1: Process timeline for the Scientific Excellence category

2.2 Newcomer category

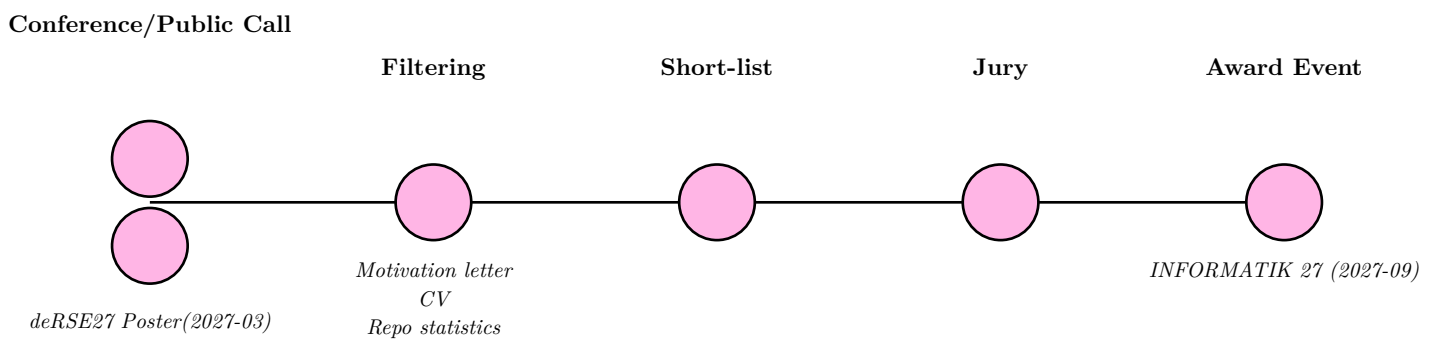


Figure 2: Process timeline for the Newcomer category

3 Filtering

Category	Filter Criteria	Filter Indicators
Artefact Track	-	-
Scientific Excellence	Scientific societies	
	Research impact	Number of citations, Number of paper downloads
		Ranking of the journal
Newcomer	Motivation letter	Relevance of RS for community
		CV
	Community Engagement, Repo statistics	Stars, Forks, Watch
		Contributor growth, Pull requests
		Issues, Commit frequency

Table 2: Filter system

3.1 Scientific societies

The general idea for involving scientific societies is to get a high quality evaluation of the *research impact* criterion by domain experts and to filter down the RS submissions.

The [Helmholtz Software Award \(p. 2\)](#) defines *scientific impact* as:

“This criterion evaluates the scientific impact of the software. Research software is regarded as high-impact if it demonstrably contributes substantial value to scientific research. A software has scientific impact if it e. g. enhances the analytic capabilities of research, or makes possible efficiency gains of research projects, or makes possible tackling new research questions. This is demonstrated by citations in in high quality publications or awards.”

And it uses the following indicators: *Narratives explaining impact, quality and number of citations demonstrating impact, awards.*

Example questions we could address to the scientific societies to measure the *research impact* of a RS:

- Does the RS solve an important problem that has not been solved before or it solves the problem in a new, innovative way?

- Does the RS increase the existing knowledge within the research field?
- Does the RS make it possible to tackle new research questions?
- How high is the re-use potential of the RS for other researchers to make valuable contributions to the research field in the future?
- Does the RS make efficiency gains of research projects possible?

Each question would be answered on an ordinal scale, eg from 1 (minimum) to 5 (outstanding).

The scientific society gives a brief justification for each questions' score and an overall assessment for each evaluated RS in the form of a free text.

On top, the following quantitative indicators are also collected to measure the *research impact*:

- *number of citations*
- *ranking of the journal*
- *number of paper downloads*

For each submitted RS the date of the collected indicators must be the same, eg the start of the awards' application period. These indicators could be provided (or not) to the scientific societies for the evaluation.

4 Jury composition & award ceremony

4.1 Jury composition

- “Scientific Excellence” category: Probably 1 jury for each sub-category (4 in total) -> domain knowledge important
- “Newcomer” category: a sample of the jury people from “Scientific Excellence” category
- How many people? Decision rules?
- Using evaluation system provided by us (see below)

List of relevant people for jury:

- [GI RSE Fachgruppe Leitungsgremium](#)
- deRSE conference orga team; see [deRSE26](#)

- [CORSES](#) — Collaborative RSE Services project
- [find.software](#) project
- [Ben Hermann](#), mentioned in a previous AK meeting

4.2 Award ceremony

- idea: INFORMATIK 27 (2027-09)
- invite someone from DFG; see [DFG Förderprogramm “Forschungssoftwareinfrastrukturen”](#)

5 Evaluation criteria and indicators

5.1 Artefact Track and Scientific Excellence

	Software Engineering Level	Research Impact	Community Engagement	FAIRness & (Reproducibility)	Maintainability & Sustainability
Definition	This criterion evaluates the adherence to best practices of coding and software engineering principles.	This criterion evaluates the scientific impact of the research software.	This criterion evaluates the degree of community engagement to the research software.	This criterion evaluates the adherence to the FAIR4RS principles, which are part of the Open Science concept.	This criterion evaluates the long-term stability and maintainability of the research software.
Indicators	<i>Best SE practices (e.g., modularity, readability, efficiency); Software has tests (unit, integration, system tests); Software test coverage (code, branch, threshold, completeness); Human code review requirement (pull requests)</i>	<i>Number of citations; Ranking of the journal; Number of paper downloads</i>	<i>Repo contribution stats (guidelines, contributors, pull requests, commit frequency, issues); Repo popularity stats (stars, watch, forks, downloads); Active communication channels and documentation pages</i>	<i>CodeMeta completeness; descriptive metadata; license; persistent identifier; archived in Software Heritage/scholarly repo; uses citation; versioning standards; containerized</i>	<i>Repo is active; has releases; up-to-date metadata; has dependency management solution; uses issue tracking system; current phase in the Software Development Life Cycle</i>
Sources	Projektantrag, EVERSE indicators , Helmholtz Software Award				

Table 3: Evaluation criteria and indicators for the Artefact Track and Scientific Excellence category

Each criterion is rated on an ordinal scale, from 1 (minimum) to 5 (outstanding).

5.2 Newcomer

	Community Engagement	
Definition		
Indicators		
Sources		

Table 4: Evaluation criteria and indicators for the Newcomer category

Each criterion is rated on an ordinal scale, from 1 (minimum) to 5 (outstanding).

Appendix